

FIRST PERSON

First person – Justin Varholick

First Person is a series of interviews with the first authors of a selection of papers published in Biology Open, helping researchers promote themselves alongside their papers. Justin Varholick is first author on 'Older spiny mice (*Acomys cahirinus*) have delayed and spatially heterogeneous ear wound regeneration', published in BIO. Justin is a postdoctoral associate in the lab of Malcolm Maden at the University of Florida, investigating environmental and physiological contributions to the plasticity of tissue regeneration.

Describe your scientific journey and your current research focus

I have always been interested in understanding how people and animals follow different developmental trajectories. It started when I was diagnosed with amblyopia – or lazy eye – when I was five and the doctors said it was because I slept with a night light on. Amblyopia is relatively inconsequential to my health, but it makes me pretty terrible at sports, which affected my friend groups and interests in childhood.

When I went to college, I learned about development and how small environmental perturbations can alter it. I eventually earned a doctorate studying how social dominance relationships contribute to individual-specific social environments, leading to individual differences in behavior and physiology. I've always read widely, and when I read about spiny mice, I had so many questions about the plasticity of their regeneration or how factors could redirect their regeneration trajectory. This is my current research focus. I think the processes underlying regeneration are quite plastic and sensitive to perturbation. Thus, we can manipulate these factors to increase or decrease regeneration quality.

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Who or what inspired you to become a scientist?

I think the doctors who tried to treat my amblyopia with a surgery that didn't reverse my amblyopia inspired me to become a scientist. Doctors use many treatments to address medical issues, but ultimately, they are limited by our current knowledge. It's up to scientists to discover more about biology to develop better treatments.

Spiny mice have a remarkable ability – almost a superpower – to regenerate their skin, hair follicles, ear cartilage, muscle, and spinal cord after injury without scarring

How would you explain the main finding of your paper?

Spiny mice have a remarkable ability – almost a superpower – to regenerate their skin, hair follicles, ear cartilage, muscle, and spinal cord after injury without scarring. However, the time it takes them



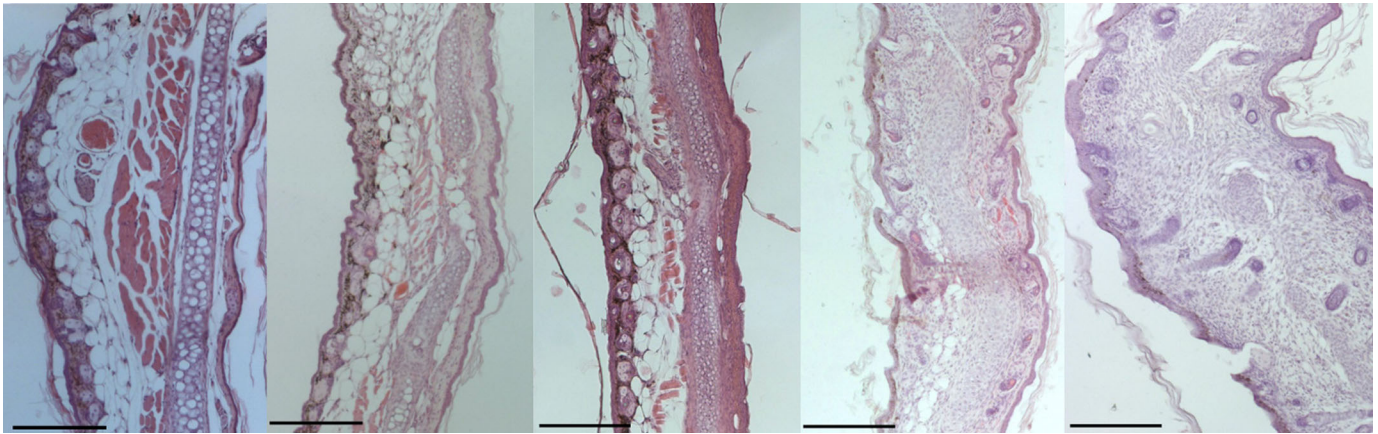
Justin Varholick

to regenerate their ear varies from 20 to 90 days. We determined that older spiny mice take longer to regenerate and regenerate less cartilage and fat cells than younger spiny mice. We thought this difference would be driven by differences in age-related nerve degeneration (i.e. peripheral neuropathy), but we found little evidence for this.

What are the potential implications of this finding for your field of research?

The direct implication is that scientists studying ear regeneration in spiny mice must consider the animal's age and the ear tissue's spatial heterogeneity in their experiment. Further experiments may delineate what factors contribute to faster regeneration with more

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Histology images showing the differences in regeneration quality across ages in spiny mice. The leftmost image shows uninjured ear tissue. Subsequent images show regenerated tissue from different aged animals increasing in age from 3–4 months old to 4–6 months, 6–9 months, and ≥ 22 months. The cartilage strip in the middle is the most apparent change, followed by the decline of fat cells. Scale bars: 250 μm .

cartilage in younger animals. This may then be used to improve regeneration in the lab and clinic.

Which part of this research project was the most rewarding?

It was fun discovering that the quality of the regenerated ear started to decline in the most distal area of the injury (i.e. furthest from the head). Because I was interested in individual differences, we made a big effort to section the entirety of the regenerated tissue. Then, under the microscope, slide by slide, we discovered there was less cartilage, muscle, and fat cells in the distal area. It was tiresome but rewarding work.

What do you enjoy most about being an early-career researcher?

I come to the lab every day to dream about the future impacts of my work and to experience the elation of discovering something new.

What piece of advice would you give to the next generation of researchers?

Follow your intuition. Your personal experiences have shaped how you view the world; you are the only person with your perspective. Science needs your perspective. It thrives off innovation and discovery. Do something new and unexpected. Think different and think big.

What's next for you?

I plan to continue studying how to manipulate regeneration quality in spiny mice and have big plans for incorporating behavior into tissue regeneration.

Reference

Varholick, J. A., Thermolice, J., Godinez, G., Dos Santos, V., Kondapaneni, R. and Maden, M. (2024). Older spiny mice (*Acomys cahirinus*) have delayed and spatially heterogeneous ear wound regeneration. *Biology Open*. **13**, bio060565. doi:10.1242/bio.060565